Typed Epistemic State under Partial Knowledge: Six Matched-Information Mechanism Studies

Anonymous authors
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Abstract

Research systems act on more than world facts: they carry partial knowledge about feasibility, failed interfaces, uncertain costs, transported certificates, unresolved questions, and representation edits. We ask whether explicitly typing and scoping that epistemic state changes decisions when competing mechanisms receive the same visible information. Six prospectively frozen exact-synthetic studies isolate one state operation at a time. Type-conditioned value-of-information planning recovers 71% of oracle utility and exceeds the identical uniform-prior planner. Scope-bound receipt reopening avoids wasteful global reopening by a wide paired margin; its advantage over never reopening at all is not separated from zero at this sample size and we do not claim it. Verification targeted at interval-dominance ambiguity yields 0.1096 mean regret versus 0.2518 for random verification at the same budget. Full-chain transport checking detects all constructed laundering cases with recall 1.000 and false-positive rate 0.000, including 68 deep splices that defeat last-hop inspection. Decision-coupled experiment selection avoids max-entropy decoys that consume 36.6% of pure-information-gain probes. Typed remint/transport improves utility when transport is available and ties re-derivation exactly in a prespecified regime where it has nothing to add. Two additional studies delimit the claim: an ideal value-of-information donor exactly closes the N1-C allocation-policy residual, and a crossover predictor is donor-absorbed on its original N2-F5B world, surviving only under misspecification. The evidence establishes mechanism isolation on frozen enumerable worlds, not deployment performance. LLM-labeled controls are deterministic proxies; chain identifiers are not cryptographic security; and no real research-interface or impossibility claim is made.

Keywords: epistemic state; partial knowledge; value of information; provenance; verification; active experimentation.

1 Introduction

A long-running research system accumulates facts about its own knowledge state: an interface is known feasible, unknown, or previously failed under a particular representation; a cost is measured or only bounded; a certificate was minted under an earlier context; an unanswered question is uncertain but may or may not affect the current decision. Flattening these states into generic booleans or confidence scores can erase information about when evidence remains applicable.

The scientific difficulty is attribution. If a typed system has more information than an untyped baseline, a positive result says little about typing itself. This study therefore uses matched-information worlds. Inside each family, every non-oracle arm receives the same serialized visible state and differs only in how it interprets or acts on that state. The strongest comparison gets first right of refusal, and each world contains a hostile regime designed to expose the shortcut that the proposed typing is meant to prevent.

Six families cover complementary operations: priors over unknown feasibility, scope-bound invalidation of failure receipts, verification under interval uncertainty, provenance-chain transport, decision-coupled experiment selection, and remint/transport across representation changes. Their purpose is not realism. Exact-synthetic construction lets the mechanism be isolated against enumerable truth, matched information, and paired outcomes.

The paper also includes negative evidence by design. One typed-state family loses its policy residual to an ideal VOI donor, and another candidate is donor-absorbed on its original world. These results matter because a mechanism-isolation programme that only publishes wins cannot establish that its first-refusal rules are doing work.

2 Methods

2.1 Prospective freezing and paired construction

Each study freezes its world generator, seed, arms, metrics, gates, tie-breaks, and honest terminal vocabulary before result-bearing execution. Episodes are paired where relevant so competing arms face the same underlying realization.

2.2 Matched information

Within a family, every non-oracle arm receives the identical serialized visible state. Mechanism comparisons therefore change the decision rule, not the available facts. Oracle arms are included only as upper/reference bounds and are not described as deployable baselines.

2.3 First right of refusal

The strongest donor-complete comparator appropriate to each construction is registered in advance. A donor tie or win closes the mechanism/policy claim. This rule is responsible for the N1-C and original-world N2-F5B negatives reported below.

2.4 Hostile validity gates

Worlds include a regime that should punish the shortcut under test: blind optimism, global reopening, partial chain inspection, decision-irrelevant information gain, stale carry-forward, or a typed operation that should provide no value. The expected trap must actually bite where prespecified; otherwise the world terminates invalid rather than yielding a positive mechanism claim.

2.5 Determinism and authority

The N4 studies use frozen seeds and deterministic decision paths. Canonical receipts encode result and scope. Arms labelled as language-model proxies refer to fixed heuristics, not to measurements of any deployed model. Authority strings explicitly limit the results to exact-synthetic mechanism isolation.

3 Results

3.1 Typed priors for unknown feasibility

In N4-A, type-conditioned priors are the only difference between the typed value-of-information arm and its uniform-prior ablation. Over 300 paired episodes, the typed arm has mean utility 3.291 versus 2.180 for uniform VOI and 4.612 for the full oracle, recovering about 71% of oracle utility. Blind optimistic commitment is driven to -13.619 mean utility and succeeds in only 19% of episodes, satisfying the hostile-world validity gate.

3.2 Scope-bound failure-receipt reopening

N4-B separates changes that invalidate a failure receipt from irrelevant context churn. Pooled mean utility is 3.199 for scoped reopening, 2.782 for never reopening, -7.813 for unscoped-change reopening, and -9.225

for always reopening. In the deliberately wasteful regime, always reopening collapses to about −13.406 while never reopening remains positive; the scoped mechanism retains the no-giveback property required by the protocol.

The scoped-versus-never gap is not a supported result, and we state this explicitly because the pooled means invite the opposite reading. Under the paired analysis of Table 1, scoped reopening is not separated from never reopening in either regime: the paired mean difference is 0.060 with 95 % interval $[-0.540, 0.634]$ in the wasteful regime and 0.774 with interval $[-0.663, 2.254]$ in the stale regime, on 200 pairs each. Both intervals contain zero. The wasteful-regime comparison is additionally 79 % ties, so the pooled means are separated mostly by a small minority of discordant episodes. What N4-B does establish at this sample size is the scoped-versus-*unscoped* contrast, where both intervals exclude zero by a wide margin ($[13.624, 16.451]$ and $[5.740, 8.255]$). The mechanism claim this study supports is therefore that scoping reopening beats reopening on unscoped change — not that reopening under scope beats never reopening at all.

3.3 Targeted verification under interval costs

N4-C gives every active arm the same budget of four edge verifications. Interval-dominance targeting yields mean scalarized regret 0.1096, compared with 0.2518 for random verification, 0.2621 for midpoint optimization without targeted verification, 0.7755 for worst-case optimization, and 1.2679 for best-case optimization. The benefit is therefore attributed to where verification is spent, not verification volume.

3.4 Full-chain certificate transport

N4-D contains 200 honest and 200 laundering chains. The full-chain checker has recall 1.000 and false-positive rate 0.000. It detects all missing-receipt, spoofed-summary, and 68 deep-splice attacks. Last-hop inspection has zero recall on the deep-splice class. This is a construction-level provenance result; the identifiers are seeded values rather than cryptographic digests.

3.5 Decision-coupled active experiments

In N4-E, pure information gain is attracted by high-entropy facts that are irrelevant to the downstream choice. It spends 36.6% of its probes on decoys and obtains mean utility 7.121, while the decision-coupled selector spends none on decoys and reaches 9.266 under the shared stopping rule. The decoy-attraction gate is part of world validity.

3.6 Remint and transport across edits

N4-F3 compares typed remint/transport with naive carry-forward and matched-budget re-derivation. In the stale-hostile regime, naive carry-forward fails on 99.5% of episodes and has mean utility about −15.916; typed transport reaches 9.421 versus 7.157 for re-derivation. In the prespecified remint-unnecessary regime, all relevant arms tie exactly at 11.809659685355605 and the typed arm spends zero remints. A positive advantage there would have invalidated the study.

3.7 Paired uncertainty for every comparison

The means reported above are point estimates. Table 1 gives the paired 95 % percentile bootstrap interval, the pair count, and the win/tie/loss decomposition for each of the twelve N4 comparisons, transcribed from the committed publication analysis. Three readings matter for how far these results should be taken.

First, two comparisons are undetermined: both N4-B scoped-versus-never contrasts have intervals containing zero, as discussed above. Every other non-degenerate comparison excludes zero.

Second, several intervals that exclude zero rest on a minority of discordant pairs, and the tie column is the honest measure of how often the mechanism changes anything at all. N4-A typed-versus-uniform VOI wins 29 % of pairs against 13 % losses with 58 % ties; N4-C wins 26 % against 7 % with 67 % ties. Both differences are consistent in direction, but they are carried by roughly a third of episodes, not by a uniform shift. The

Table 1: Paired uncertainty for every N4 comparison in the committed publication analysis. Intervals are 95 % percentile bootstrap intervals over paired episode differences (5000 deterministic resamples, seeds recorded per row in the source artifact). Win/tie/loss is the paired decomposition over the same n pairs. Intervals are per-comparison and are *not* adjusted for the twelve comparisons shown; any family-wise adjustment widens them, so rows already containing zero would continue to contain zero. Rows marked † have an interval containing zero: the contrast is *undetermined* at this n and is not a positive result. The row marked ‡ is the prespecified remint-unnecessary regime, where the exact tie on all pairs is the passing outcome and a positive advantage would have invalidated the study; its interval is $[0, 0]$ by construction and is not an undetermined contrast.

Study	Comparison	Mean diff.	95 % CI	n	win/tie/loss
N4-A	typed VOI vs. greedy known-graph	2.933	[2.556, 3.301]	300	0.55/0.30/0.15
N4-A	typed VOI vs. uniform-prior VOI	1.111	[0.833, 1.400]	300	0.29/0.58/0.13
N4-B	scoped vs. never reopen (wasteful)†	0.060	[−0.540, 0.634]	200	0.17/0.79/0.04
N4-B	scoped vs. unscoped reopen (wasteful)	15.050	[13.624, 16.451]	200	0.78/0.11/0.11
N4-B	scoped vs. never reopen (stale)†	0.774	[−0.663, 2.254]	200	0.59/0.20/0.21
N4-B	scoped vs. unscoped reopen (stale)	6.973	[5.740, 8.255]	200	0.52/0.35/0.13
N4-C	interval-Pareto vs. random verification	0.142	[0.100, 0.187]	400	0.26/0.67/0.07
N4-E	decision VOI vs. information gain	2.146	[1.976, 2.299]	400	0.95/0.03/0.02
N4-E	decision VOI vs. LLM proxy	0.277	[0.119, 0.412]	400	0.45/0.44/0.14
N4-F3	typed transport vs. naive carry-forward	17.242	[15.577, 18.836]	200	0.69/0.30/0.00
N4-F3	typed transport vs. re-derivation (mixed)	2.264	[1.717, 2.825]	200	0.32/0.69/0.00
N4-F3	typed transport vs. re-derivation (no remint needed)‡	0.000	[0.000, 0.000]	200	0.00/1.00/0.00

Metric note: N4-A, N4-E and N4-F3 differences are in episode utility; N4-B in mean round utility; N4-C in scalarized regret reduction. N4-D is a frozen exact census (200 laundering / 200 honest chains), not a sampled paired comparison, and therefore has no interval; its counts are reported in the text.

large-margin results — N4-E versus information gain (95 % wins) and N4-F3 versus naive carry-forward (69 % wins, zero losses) — are the ones that hold on most episodes.

Third, the intervals are per-comparison and unadjusted for multiplicity. Twelve comparisons are reported, nine of which exclude zero. Any family-wise adjustment widens all of them. The one direction that follows without further computation is that the two undetermined contrasts can only remain undetermined. We do not assert here which of the nine would survive adjustment. The committed artifact stores summary statistics rather than the paired difference vectors, so corrected *bootstrap intervals on the mean* cannot be derived from it. A family-wise-correctable exact test is available by a different route, over the win/loss counts rather than the means, and is reported separately; we keep the unadjusted intervals here because the mean is the registered estimand for these comparisons. Comparisons whose interval lies close to zero relative to its width — N4-C and N4-E against the LLM proxy — are the ones a family-wise correction would put at risk.

3.8 Negative first-refusal cases

N1-C finds a typed/scoped failure-state benefit over a scoping ablation, but an ideal VOI allocator given the same typed facts matches the candidate policy exactly at solve rate 0.9866. The allocation-policy residual is therefore closed. N2-F5B similarly gives a model-selection donor first refusal: candidate and donor both score 0.9948 on the original world, leaving only the narrower misspecified-world edge (0.9844 versus 0.9531) as a surviving bounded result.

4 Discussion

Across the six primary families, the common pattern is conditional rather than universal. Typing/scoping matters when it changes which uncertainty, receipt, certificate, or experiment is relevant to the decision. The matched-information ablations make that interpretation stronger than a comparison in which the typed arm simply sees richer state.

Equally important is the no-value behavior. N4-F3's exact tie shows that the mechanism does not need to win everywhere to support its intended claim; in the regime where remint/transport has no informational value, it does nothing and ties. N1-C pushes the same principle further: typed state can matter while the proposed policy adds nothing beyond an ideal donor. The method claim should stop at the state representation in that case.

Hostile controls are useful because they test whether a synthetic world actually exercises the proposed failure mode. They do not make the world realistic. A model can perform well on all six constructions and still fail on real research interfaces for reasons absent from the generators: mis-specified state types, noisy provenance, nonstationary costs, strategic adversaries, or poor semantic extraction from raw evidence.

The next scientifically discriminating step is therefore transfer, not more synthetic variants of the same traps. A real-domain instantiation would need independently auditable state labels and matched-information baselines rather than silently substituting a richer system for the typed mechanism.

5 Related work and boundary

The component mechanisms have established parents. Robust optimization studies decision making under uncertain inputs and the trade-off between nominal performance and protection against uncertainty Bertsimas & Sim (2004). Data-provenance work formalizes where derived records originate and which source data influence them Buneman et al. (2001), while active-learning literature studies how to choose information-gathering actions rather than passively consume labels Settles (2009). These ingredients are not claimed individually.

Recent agent work makes the boundary substantially tighter than those classical parents alone. STALE directly benchmarks whether agents recognize and act on invalidated memory Chao et al. (2026). MAP-Graph uses a typed provenance graph, ancestry, permission filtering, path trust, and action gating as an operational memory control Wang et al. (2026), and provenance-sensitivity auditing varies source authority while holding other task factors fixed Liao (2026). Long-horizon studies also show that governance constraints can be erased by context compaction Chen (2026) and that omission and commission constraints can decay differently under context pressure Gamage (2026). We therefore do not claim priority on typed memory, provenance-sensitive action, stale-state revision, or governed/versioned memory.

The closest new parent is a budgeted-verification study of inherited stale constraints Nakayashiki (2026). It fixes the verification budget at two records in every arm and includes a random-record control, so matched-budget verification against random verification is explicitly *not* an ORION-08 novelty claim. Its object is sampled behavioral effect under a fixed allocation. ORION-08's residual object is different: an exact finite decision-sufficiency problem. The binding-sufficiency lattice states when all worlds merged by a readable binding share a common optimal action and hence when deterministic zero regret is possible; refinement is useful only when it separates a fibre whose worlds do not share an optimal action. The synthetic studies then instantiate that mechanism under matched information and prespecified hostile or no-value controls.

This exact-versus-sampled distinction should not be read as an external-validity claim. The recent budgeted-verification result supplies an informative independent neighboring experiment, and a separate transfer note records that its reported large-effect and no-effect conditions are consistent with the fibre criterion. That note is a prediction-level mapping, not an episode-level re-analysis; a released-data fibre mapping that reverses the predicted effect/null ordering would falsify it.

The bounded candidate contribution is therefore the common mechanism contract across six separately frozen exact-synthetic partial-knowledge operations: hold the factual world fixed, vary the explicit epistemic binding available to the decision rule, use a matched comparator, include hostile or no-value controls, and report an exact bounded terminal. The cross-family taxonomy is a post-study synthesis. It does not establish deployment prevalence or real-domain superiority.

This paper is separate from the negative-recovery methodology reported in companion work, which asks how research outcomes are frozen, retained, and reopened; ORION-08 asks whether particular types and scopes of epistemic state are load-bearing inside controlled decision problems.

Ignorance is already known not to be one thing. The premise that epistemic state needs typing rather than a single not-known flag has independent formal support. Kubyshkina and Petrolo give sound and complete systems for three distinct notions — ignorance whether, ignorance as unknown truth, and disbelieving ignorance, in which the agent holds a false belief about a true proposition Kubyshkina & Petrolo (2026). The distinctions are load-bearing rather than terminological: disbelieving ignorance is not interdefinable with knowledge on the relevant frames, so a system that records only *not known* cannot express it. Their topic-sensitive semantics adds a further separation, since an agent may be ignorant of one of two logically equivalent propositions when she does not grasp the topic of the other.

That literature settles the conceptual question this paper would otherwise have to argue, and settles it more rigorously than an empirical study could: the types are formally distinct, and collapsing them loses expressive power. What it does not address is whether the distinction is *observable* in a running system. An axiomatization tells you the states are different in principle; it does not tell you whether two mechanisms that are supposed to occupy different epistemic states behave differently under matched information, or whether an implementation that claims to distinguish them actually does. That is the question the mechanism studies here are built to answer, and it is why the information available to each arm is matched rather than merely reported.

6 Limitations

All primary evidence is exact synthetic. The generators provide enumerable truth and cleanly typed state, which is the source of internal control and also the main external-validity limitation. Real research systems must infer whether a failure is stale, whether a certificate's scope changed, and whether an uncertainty is decision-relevant.

The LLM-labeled baselines are deterministic proxies. No conclusion about current or future language models follows. N4-D does not study cryptographic attackers; its hashes are construction identifiers and its completeness statement assumes end-to-end receipts cannot be forged consistently within the world definition.

N4-C reports scalarized regret, not full Pareto-front quality. N4-B excludes intra-episode receipt accrual, and two of its comparisons are underpowered rather than negative: scoped versus never reopening has a paired interval containing zero in both regimes (Table 1), so the study is not entitled to a scoped-versus-never claim in either direction. The programme has no lower-bound/impossibility theorem showing that typing is necessary for every mechanism family, and the registered real/generic-LLM comparison rung was not executed.

Reported intervals are per-comparison. Twelve N4 comparisons are reported without a family-wise correction. The committed publication analysis stores summary statistics rather than the paired difference vectors, so corrected bootstrap intervals on the mean cannot be recomputed from it; an exact test over the win/loss counts is correctable and is reported separately, but it addresses the direction of the paired difference rather than the registered mean estimand, and we do not treat it as licensing a claim the mean intervals do not support. Comparisons whose interval sits close to zero relative to its width — N4-C, and N4-E against the LLM proxy — should be read as the ones most exposed to that correction.

Finally, multiple positive synthetic families do not constitute independent populations. Their common design discipline reduces some confounds but cannot support a statistical prevalence claim such as 'typing helps in six out of six kinds of real research problem'.

7 Conclusion

On six frozen partial-knowledge constructions, explicitly typed and scoped epistemic state changes decision quality even when compared mechanisms receive the same visible information. The effect appears in priors, receipt invalidation, verification targeting, provenance-chain checking, experiment choice, and remint/transport. Prespecified hostile controls show that the corresponding untyped shortcuts are genuinely exercised, while an exact no-value tie and two donor-closure cases bound the claim.

The result is a mechanism-isolation benchmark, not a deployment result. It supports studying epistemic-state type and scope as first-class decision variables and provides deterministic worlds in which future mechanisms can be compared without conflating state content with state interpretation.

8 Reproducibility

Protocols are committed under `development/orion-q-nlane-closure/`; canonical results and generating modules are under `research/extensions/orion-q/nlanes/`. The replay ledger records deterministic re-execution of the core N-lane receipts. Each family uses a frozen seed, explicit gates, and a terminal vocabulary that distinguishes mechanism failure from world invalidity.

A reproduction should first verify the protocol/result binding, then rerun the world and all registered arms, then recompute the headline matched comparator and hostile validity gate. The negative N1-C and N2-F5B families must be verified with equal prominence. Paper-level verification steps and non-equivalences are listed in `REPRODUCE.md`.

9 Ethics, safety and resource statement

No human participants, animals, clinical data, or personal data are used. Because the studies construct hostile failure modes, the principal reporting risk is over-interpreting those constructions as security or deployment evidence. The manuscript therefore labels them as exact-synthetic mechanism tests and keeps cryptographic, real-adversary, and real-agent claims outside the authority boundary.

Computational work is deterministic and bounded by the frozen episode counts/enumerations. Negative and donor-absorbed results are retained rather than discarded. No result here can self-authorize a promotion to a broader scientific claim, a novelty claim, a deployment decision, or any physical quantum claim.

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